

Ryan Tremarco

Senior Software Engineer

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Experience

Vet-AI

Senior Backend Engineer: November 2024 - Present, Remote

- Developing Go APIs and using Encore to manage infrastructure orchestration.
- Integrating with GetStream and their webhook system to build a chat-focused variant of online veterinary consultations.
- Creating and integrating clinically-validated B2C veterinary AI agents into new Chat offering using Langchain in Python and Gemini as a base model.
- Built specialised RAG for triaging customers pre-vet consultation, leveraging business-specific knowledge bases and cutting down consultation times.
- Integrating custom-built Vertex AI predictive models for analysing user-submitted images and videos to assist in automated triage in Symptom Checker product.

Benefex

Senior Backend Engineer: May 2023 - September 2024, Remote / Southampton

Backend Engineer: January 2023 - April 2024, Remote / Southampton

- Led the development of a Microsoft Teams application integrating with core products for notification delivery and product interaction.
- Developed event-driven Go microservices, deployed to GCP kubernetes clusters.
- Owned and built a centralised notifications system for the product suite, enabling email, web, mobile, and Microsoft Teams notifications to users from our products.
- Led the replacement of ageing parts of the system, maintaining existing contracts and enabling customer globalisation (an average revenue increase of 33% per customer).
- Proposed and built shared standard error, event outboxing, and API client Go packages.

VISFO

Backend Engineer: July 2021 - January 2023, Remote / Brighthouse

- Technical ownership of 'Curator' project to replace and then enhance process for progressing and providing feedback on employee-submitted work.
- Developed Golang API microservices following MVC patterns hosted on GCP.
- Implemented gateway aggregation using GraphQL to prevent over-fetching.
- Built dataloader for GraphQL to batch db and network calls, greatly reducing load.
- Built and managed ETL/ELT pipelines using python, Airbyte, and DBT.
- Implemented Agile practices and maintained a guiding role at ceremonies.

Her Majesty's Government

Full Stack Engineer: September 2017 - July 2021, Manchester

- Development of Java REST microservices using Spring MVC and Eureka.
- Development of Node JS APIs using a custom fork of PKI auth-enabled Express.
- Deploying services to OpenShift using S2I to AWS EKS.
- Development of React apps; hands-on UX/UI using customer-centric workshops.
- Maintenance and expansion of data pipelines built in Apache Camel.

About Me

Hello! I'm a Software Engineer with experience working across the full stack of web development and I have a particular liking to backend API development.

I do my best to bring a level of pragmatism to the projects that I am a part of. I understand and make as much use of best practice as possible, but I also know that chasing perfection in our work is a never-ending task. Knowing where to stop, or where best to break a convention in order to deliver projects is as great a part of being an Engineer as system design or implementation in my opinion.

Communication is also something that is very important to me and I have worked hard to improve over the years. I like to share information as much as possible with colleagues to help reduce single-point-of-failure risks and siloing within teams. To this end I have tried to make it a habit to organise pairing / mobbing / learning / lightning talk sessions with team members in order to share knowledge and help upskill others in areas that I am a local expert in.

At home I like to experiment with tech that I don't get to use day-to-day like Rust and Haskell. I have found that I personally learn best by doing, so I like to take part in programming events such as Advent of Code in order to help with this. More recently I've been dabbling in game development via game jams and I'm particularly enjoying using Godot's node system.

Outside of tech, my interests generally lean towards hobbies like video games, Warhammer, tabletop RPGs like Mothership and D&D, and boardgames. I also dabble in language learning and have been teaching myself Japanese for a few years now; I'm hoping to pick up some Italian in the future too.

Skills

Languages:

- Go
- Python
- Java + Spring
- JavaScript + React
- Bash
- Rust

Tooling:

- Encore
- Langchain
- Vertex AI
- Linux / WSL2
- Git
- Grafana + Prometheus
- PostgreSQL
- MongoDB
- MariaDB

Infrastructure:

- GCP
- OpenShift
- AWS
- Terraform

Education

University of Roehampton / BSc Digital and Technology Solutions (1st)
SEPTEMBER 2017 - JANUARY 2021

While working at HMG, I was fortunate enough to be sponsored to complete my first-class degree with honours. The work for this was completed in my own time alongside full-time software projects.